Devesh Khemani

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British Citizen | Eligible for UK Security Clearance (SC)

Education

Imperial College London — London, UK
Electrical and Electronic Engineering BEng | Predicted: 2:1 — 2025–2028
Relevant Modules: Control Systems, Signal and Systems, Communications, Electromagnetism, Circuits and Systems

King Edward VI Camp Hill Grammar School for Boys — Birmingham, UK
*A*A*A* in Mathematics, Physics, Computer Science A-Level* — 2024

Technical and Research Projects

Mixed-Signal Electro-Optical LoS Tracker & Inertially Stabilised Platform — Jun 2026 – Sep 2026
MPU-6050 IMU, Complementary Filtering, PID Control, Embedded C++, STM32F4, Simulink, KiCad, LTspice

- Engineered an inertially stabilised platform, using a 200 Hz C++ complementary filter to fuse MPU-6050 gyroscope and accelerometer data.
- Modeled pan-tilt servomechanism dynamics in Simulink, tuning a discrete-time PID loop with a 1 ms sample period to optimise actuator step response, reducing settling time by 35% against a Ziegler-Nichols baseline while maintaining <5% overshoot.
- Applied IMU angular-rate feed-forward within a model-in-the-loop (MIL) simulation, integrating the tuned PID model with simulated wearer kinematics to evaluate dynamic tracking-lag suppression and disturbance rejection.
- Engineered the final PID controller in C++ and verified closed-loop convergence within a software-in-the-loop (SIL) simulation, incorporating simulated sensor noise and target dynamics to achieve $<3.3^{\circ}$ closed-loop Line-of-Sight tracking accuracy.
- Designed a 4-channel optical front-end in LTspice to process photodiode signals generated by a transmitted 38 kHz modulated IR carrier.
- Simulated 73 dB attenuation of 100 Hz ambient interference using a 38 kHz 4th-order Sallen-Key filter integrated with an LM393 Schmitt trigger to achieve chatter-free target discrimination under solar saturation and 1–100 kHz out-of-band interference.
- Routed, hand-soldered and bench-tested a mixed-signal PCB in KiCad, isolating sensitive analog stages from PWM servo switching transients.
- Deployed the 1 kHz PID controller on an STM32F4 to process analog photodiode signals, gating 12-bit ADC amplitudes with Schmitt-triggered digital inputs for target-validity gating, utilising optical deadbanding and derivative filtering to suppress servo hunting.

Autonomous Systems & Sensing in Contested Electromagnetic Environments — Oct 2022–Sep 2023
Independent Research Project & Simulation (Python, Bayesian Sensor Fusion, NumPy)

- Analysed navigation and tracking resilience against active GPS spoofing, RF jamming and sensor degradation, benchmarking optical vs. RF/radar fusion architectures to demonstrate that multi-modal fusion is critical for target validity in contested electronic warfare conditions.
- Developed a multi-sensor Bayesian fusion simulation across thermal and optical channels in Python, modeling intrinsic and distance-scaled noise in quadrature to compute a posterior occupancy grid, reducing target state uncertainty by ∼30% compared to single-sensor modalities.
- Formulated confidence-threshold fail-safe criteria and human-in-the-loop authorisation gates to withhold autonomous actuation during low-confidence sensor states triggered by conflicting multi-modal telemetry or signal-to-noise degradation.

Teleoperated Lunar Exploration Rover — May 2026 – Jun 2026
Embedded C++, ARM Cortex-M0+, Analog Signal Conditioning, Sensor Interfacing (Hall/IR/RF), LTspice

- Co-developed a teleoperated rover for real-time rock classification using infrared, magnetic, ultrasonic, and RF sensing in a 5-person team.
- Designed and simulated analog front-ends in LTspice for 950 nm infrared and Hall-effect magnetic sensors, using active filtering to reject 100 Hz ambient-light interference and an MCP6022 gain stage, validating signal levels on an oscilloscope.
- Implemented embedded C++ routines on an ARM Cortex-M0+ to acquire 10-bit ADC telemetry, implementing a calibrated ±10-count deadband around the quiescent baseline achieving 0% false-positive target detections across 11 test trials.

Industry Exposure

Work Experience | Virgin Media O2 — July 2023

- Assessed hardware reliability and IoT sensor networks, scoping fibre-optic transmission principles to evaluate infrastructure in a team project.

Hobbies and Interests

Brazilian Jiu-Jitsu | National Level Competitor — Aug 2021 – Present

- Gold medallist: NoGi British Open, All Stars Brazilian Jiu-Jitsu Nationals, and Brazilian Jiu-Jitsu Junior National Championship

Technical Skills

Lab & Testing: Digital Storage Oscilloscopes (DSO), Function Generators, SMD Soldering/Rework, Logic Analyzers, Multimeters
Navigation & Control: Sensor Fusion (Complementary/Kalman Filtering), PID Control, IMU Interfacing, Closed-Loop Stabilisation
EDA, CAD & Tools: LTspice, KiCad 7, MATLAB, Simulink, Fusion 360, Git, Linux
Hardware & Embedded: TIAs, Sallen-Key Filters, Mixed-Signal PCB Layout, C/C++ (STM32F4, ARM Cortex-M0/M4), Python